Aryan Harkawat

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Education

University of Texas at Austin — Austin, TX
Bachelor of Science in Computer Science — *May 2028*

- **Coursework:** Operating Systems, Computer Architecture, High Performance Computing, Algorithms & Complexity, Data Structures, Linear Algebra, Statistics and Probability, Principles of Machine Learning

Work Experience

Stripe — Jan 2027 – May 2027
Incoming Software Engineer Intern

Stripe — May 2026 – Aug 2026
Software Engineer Intern, Core Infrastructure

- Rebuilt Stripe's service mesh load balancer to limit routing to overloaded backends (power-of-two-choices on CPU and queue depth), cutting overload errors 40% and saving ∼$200K/yr across 50 services at 400M+ requests/day
- Added saturation routing so overloaded backends reject excess requests early and the mesh reroutes them, spilling to neighboring zones when a whole zone runs hot, cutting p99 latency 23%
- Built a replay harness that runs production traffic through a simulated fleet; caught two regressions before rollout

Trend Micro — Aug 2025 – May 2026
Software Automation Intern

- Built Python test harness that runs 1,500 regression tests on the endpoint security agent across Windows and Linux VMs every night, cutting release testing from a week to a day and catching 30+ bugs before release
- Automated payroll reconciliation with a nightly Python pipeline that matches ∼12K records across Excel and shared-drive sources, cutting the monthly close from days to under an hour and flagging mismatches for audit

Mercor — Feb 2025 – Apr 2025
Machine Learning Intern

- Built the evaluation harness (Hugging Face Transformers) that benchmarked 6 fine-tuned BERT ticket classifiers and picked the production model, cutting a day of manual comparison to under an hour
- Quantized the production model to 8-bit with ONNX Runtime, tripling CPU throughput to ∼140 tickets/s and removing the need for GPU inference
- Wrote Pandas checks that removed duplicate and conflicting labels from 1M+ tickets, raising F1 by 2 points

Personal Projects

Concertify | *Go, gRPC, Python, PyTorch, Node.js, Redis, Prometheus, Docker* | Demo · Source

- Built a two-stage concert recommender (500+ users) that ranks Ticketmaster shows with per-user logistic regression models trained in PyTorch on Spotify listening history
- Ranked shows for groups of up to 12 in a Go gRPC service by blending average and worst-case member match
- Cut initial loading time from 9s to 2.5s under a 5 req/s API quota with Redis single-flight caching and rate limiting

Herd | *Go, Envoy, gRPC, Kubernetes, Prometheus, Docker* | Source

- Built a Go gateway for self-hosted LLMs that routes requests to servers by prompt caching and available memory, cutting p99 first-token latency 55% vs round-robin and holding 13.5s vs 292s with slow outlier servers
- Handled overload by shedding low-priority requests, fair-queuing the rest per tenant, and spilling over to a cloud API within each tenant's budget
- Added an Envoy integration modeled on the Kubernetes inference gateway standard to run inside a service mesh

Volumetric Cloud Renderer | *C++, OpenGL, GLSL, CMake* | Source

- Built real-time volumetric cloud renderer that ray-marches procedural 3D noise generated in compute shaders, with physically based light scattering and a weather map controlling cloud type and coverage
- Cut render cost 3.5× by ray-marching at half resolution with frame reuse, blending offset samples over time to match full-resolution quality

Technical Skills

Languages: Go, Java, Python, C, C++, SQL, TypeScript, JavaScript
Technologies: gRPC, Protobuf, Kubernetes, Redis, Docker, Prometheus, PyTorch, React, Node.js, AWS